HWRS 561b Homework #6

- Assigned: Monday, 19 April 2027
- Due: Thursday, 29 April 2027 at 10:00 pm MST (tentative; upload answers in PDF or Jupyter Notebook to D2L)
- Instructor: Ty Ferré (tyferre@arizona.edu)
- Teaching Assistant: Nathan Strom (nstrom@arizona.edu)
- Semester: Spring 2027

1. (100 points) PFAS transport in the vadose zone using the PFAS-LEACH Excel tool.

   Download the PFAS-LEACH Tier 3 and 4 Excel tool here. Consult the User Guide included in the repository for detailed instructions. The tool includes built-in methods for estimating model parameters, as well as databases for soils and PFAS chemical properties (e.g., K_{oc} and K_{aw}).

   For this problem, use only Module0 and the basic solver and keep the default parameters unless otherwise instructed.

   - (a) (30 points) Read the User Guide and explain the fundamental differences among the conceptual models underlying Tier 3, Tier 4, and the standard USEPA DAF approach.
   - (b) (40 points) Run the Excel tool for PFOA using default parameters, which computes the Tier 3 (PFAS-LEACH-Analytical) and Tier 4 (PFAS-LEACH-DAF) models, and the USEPA standard DAF approach.
     1. (10 points) Report the SSLs in µg/kg for Tier 3 and Tier 4. Then, re-run with air-water interfacial sorption disabled by setting $K_{aw} = 0$. Report the new SSLs and compare them with $\text{SSL}_{K_{aw} \neq 0}$. Explain the differences.
     2. (20 points) Compare the SSLs among Tier 3, Tier 4, and the standard USEPA DAF approach with and without air–water interfacial adsorption. Explain your comparisons and connect them to your answer in part (a).
   - (c) (30 points) Reduce the volumetric water content by half. Then click the "Estimate SF" and "Estimate A_{aw}" buttons to recompute the scaling factor and air-water interfacial area. Keep all other parameters unchanged, and rerun the Excel tool for PFOA.
     1. (20 points) Compare the resulting SSLs with those from part (b). Explain all of the mechanisms by which the change in water content alters the computed SSLs.
     2. (10 points) Based on your results from part (1) in (c), briefly discuss how seasonal or interannual variability in vadose zone water content affects PFAS leaching risk and the implications for long-term site monitoring strategies.